C

ORDINARY DIFFERENTIAL EQUATIONS • UNIT 2.01

Linear First-Order Equations & Integrating Factors

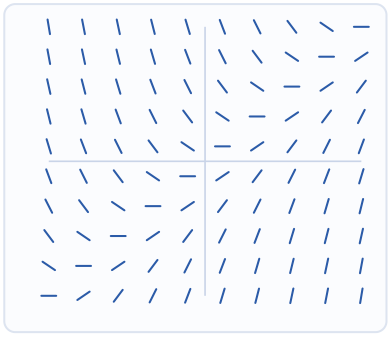
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Linear First-Order Equations & Integrating Factors Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



A 39

B 24

C 42

D 53

2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

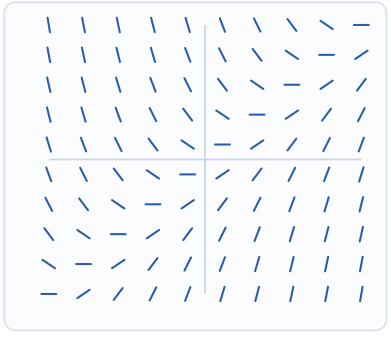
B $e^2 - 1$

C $3 - 2e^{-2}$

D $3 - 2e^{-1}$

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D 1.220

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

A 5

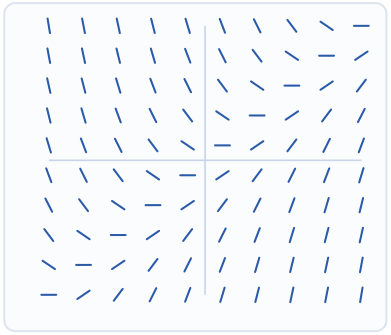
B -5

C 6

D -6

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



A 7

B 6

C 8

D 5

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

A $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

B $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

C $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

D $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 44

B 43

C 24

D 40

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

A 170

B 151

C 173

D 187

9 MCQ

The Laplace transform of $f(x) = 95e^{17x}$ is $g(s) = \dfrac{95}{s - 17}$ for $s > 17$. Determine the evaluated transform $g(18)$.

A 95

B 98

C 78

D 100

10 MCQ

Consider the first-order linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$. If the integrating factor for this equation is given by e^{x^2} and $Q(x) = 2xe^{x^2}$, find the value of the constant C in the general solution if $y(0) = 3$. The general solution is of the form $y = e^{-x^2}\left(\int Q(x)e^{-x^2}dx + C\right)$.

A 1

B 3

C 4

D 6

11 MCQ

The separable initial value problem $\dfrac{dU}{dv} = 28v$ satisfies $U(0) = 9$. Determine the solution value $U(16)$.

A 3584

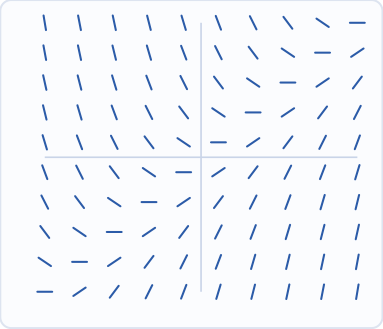
B 3607

C 3596

D 3593

12 MCQ

The separable initial value problem $\frac{dP}{dr} = 34r$ satisfies $P(0) = 8$. Determine the solution value $P(7)$.



A 858

B 844

C 841

D 833

13 MCQ

The Laplace transform of $R(m) = 48e^{20m}$ is $S(s) = \frac{48}{s - 20}$ for $s > 20$. Determine the evaluated transform $S(21)$.

A 28

B 51

C 48

D 56

14 MCQ

The second-order linear homogeneous ODE $h''(u) - 42h'(u) + 272h(u) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 45

B 50

C 23

D 42

15 MCQ

The separable initial value problem $\frac{dC}{dq} = 12q$ satisfies $C(0) = 14$. Determine the solution value $C(18)$.

A 1944

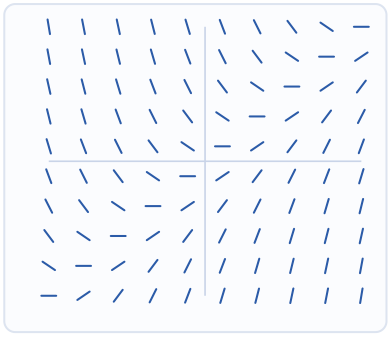
B 1958

C 1961

D 1964

16 MCQ

The Laplace transform of $E(p) = 100e^{24p}$ is $L(s) = \dfrac{100}{s-24}$ for $s > 24$. Determine the evaluated transform $L(25)$.



A 100

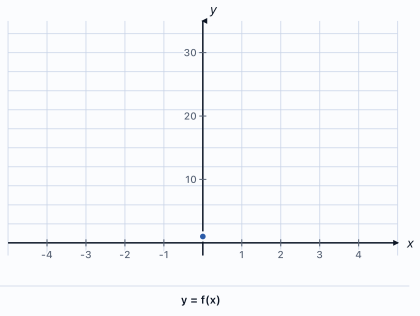
B 110

C 76

D 103

17 MCQ

Solve $y' + 3y = e^{2x}$ with $y(0) = 1$. What is the value of $y(1)$?



A $\frac{1}{5}e^2 + \frac{1}{5}e^{-3}$

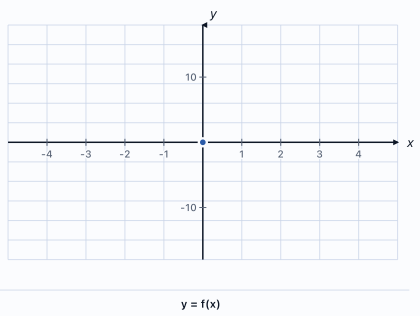
B $\frac{1}{2}e^2 + \frac{1}{2}e^{-3}$

C $e^2 + 4e^{-3}$

D $\frac{1}{5}e^2 + \frac{4}{5}e^{-3}$

18 MCQ

Solve $y' + \frac{1}{x}y = 4x$ for $x > 0$ with $y(1) = 3$. What is $y(2)$?



A $\frac{19}{3}$

B $\frac{37}{6}$

C 6

D $\frac{35}{6}$

19 MCQ

Solve the initial value problem $y' - 2y = 8$ with $y(0) = 0$. What is $y(\ln 2)$?

A 4

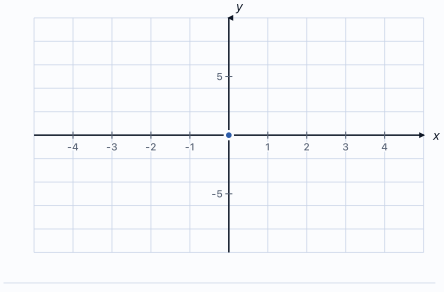
B 12

C 8

D 16

20 MCQ

Solve $y' + y = 2x$ with $y(0) = 1$. What is the value of $y(1)$?



A $2 - 3e^{-1}$

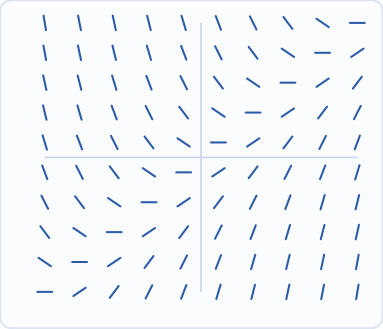
B $3e^{-1}$

C $2 + 3e^{-1}$

D e^{-1}

21 MCQ

Given $y' + 4y = 12$ with $y(0) = 7$, what is $\lim_{x \to \infty} y(x)$?



A 3

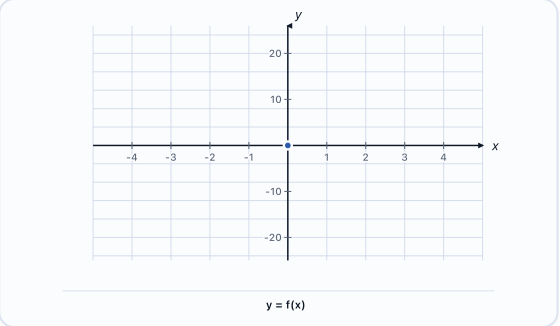
B 12

C 7

D 0

22 MCQ

Solve $xy' + 2y = 6x$ for $x > 0$, subject to $y(1) = 4$. What is $y(2)$?



A $\frac{7}{2}$

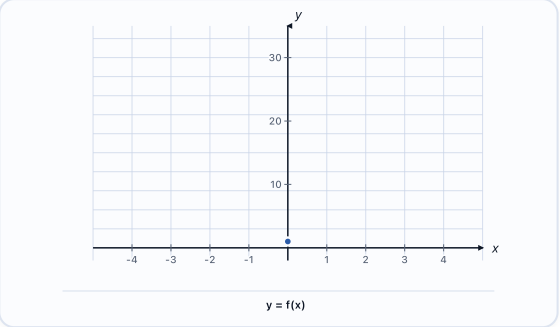
B 4

C $\frac{9}{2}$

D 5

23 MCQ

Solve $y' - y = e^x$ with $y(0) = 5$. What is the value of $y(1)$?



A $6e$

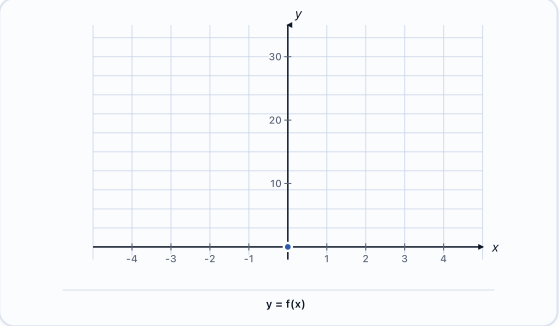
B $7e$

C $6e - 1$

D $5e$

24 MCQ

Solve $y' + 3x^2y = 6x^2$ with $y(0) = 5$. What is the value of $y(1)$?



A $2 + 5e^{-1}$

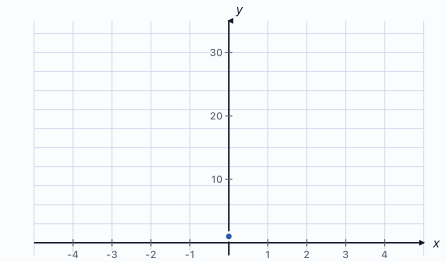
B $3 + 2e^{-1}$

C $2 - 3e^{-1}$

D $2 + 3e^{-1}$

25 MCQ

Solve $y' + 2y = e^{-2x}$ with $y(0) = 3$. What is $y(2)$?



A $3e^{-4}$

B $2e^{-4}$

C $5e^{-4}$

D $6e^{-4}$